

CSV Data Editor

Internal Tool Case Study - Python, Flask, React, PostgreSQL, Docker, AWS EC2

Backend

Internal Tools

React Integration

Data Processing

Overview

Developed and integrated an internal CSV processing tool to improve how operational teams validate, edit, and export structured CSV data. The solution was connected into the company's internal system as part of a React-based application, giving users a more reliable workflow.

Problem

- Manual CSV handling created repetitive work and increased the risk of data inconsistencies.
- Teams needed a faster, more structured way to review and correct CSV files.
- The workflow needed to fit into an existing internal system and be simple for non-technical users.

Solution

Built a web-based tool with backend logic for CSV parsing, validation, secure editing workflows, and export functionality. Integrated the tool into the internal platform through a React application/module, connecting backend data-processing features with a user-focused interface.

Key Contributions

- Created backend logic for CSV parsing, validation, editing, and export flows.
- Integrated the tool into the company's internal React-based system.
- Connected operational requirements with technical implementation.
- Supported deployment and environment setup using Git, Docker, AWS EC2, and Linux-based services.
- Collaborated with stakeholders to translate business pain points into technical requirements.

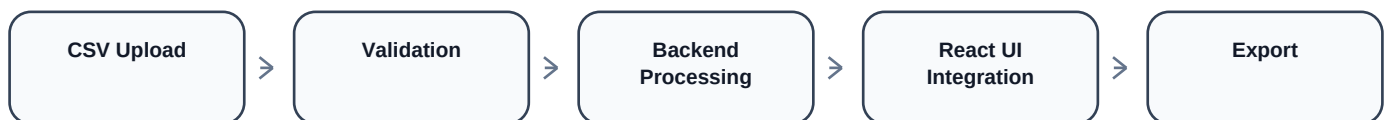
Tech Stack

Python · Flask · React · JavaScript · PostgreSQL · Docker · Git · AWS EC2 · Linux

Impact

- Reduced manual CSV processing steps for internal users.
- Improved data reliability through a structured validation and editing workflow.
- Built a practical bridge between operational processes and backend automation.

High-Level Workflow



What this project demonstrates: backend development with real business context, data-processing logic, React integration, deployment exposure, stakeholder communication, and the ability to build internal tools that solve operational problems.

Note: This case study intentionally avoids confidential company data, internal screenshots, proprietary business logic, and sensitive implementation details. It is intended for portfolio and recruiter review purposes only.

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